Solubility Equilibria

Answer key with worked solutions

A. Expressions

1. Ksp expressions.

> (a) $\mathrm{BaSO_4\,(s)} \rightleftharpoons \mathrm{Ba^{2+}} + \mathrm{SO_4^{2-}}$; $K_{sp} = [\mathrm{Ba^{2+}}][\mathrm{SO_4^{2-}}]$
> (b) $\mathrm{Mg(OH)_2\,(s)} \rightleftharpoons \mathrm{Mg^{2+}} + 2\,\mathrm{OH^-}$; $K_{sp} = [\mathrm{Mg^{2+}}][\mathrm{OH^-}]^2$
> (c) $\mathrm{Ag_2CrO_4\,(s)} \rightleftharpoons 2\,\mathrm{Ag^+} + \mathrm{CrO_4^{2-}}$; $K_{sp} = [\mathrm{Ag^+}]^2[\mathrm{CrO_4^{2-}}]$
> (d) $\mathrm{Ca_3(PO_4)_2\,(s)} \rightleftharpoons 3\,\mathrm{Ca^{2+}} + 2\,\mathrm{PO_4^{3-}}$; $K_{sp} = [\mathrm{Ca^{2+}}]^3[\mathrm{PO_4^{3-}}]^2$

B. Molar solubility

2. $\mathrm{BaSO_4}$.

> $s = \sqrt{1.1 \times 10^{-10}} = \mathbf{1.0 \times 10^{-5}}$ mol/L; 1.05×10^{-5} mol/L $\times$ 233.39 g/mol $= \mathbf{2.4 \times 10^{-3}}$ g/L

3. $\mathrm{Mg(OH)_2}$.

> (a) $K_{sp} = 4s^3$, so $s = \sqrt[3]{\dfrac{1.8 \times 10^{-11}}{4}} = \mathbf{1.7 \times 10^{-4}}$ mol/L
> (b) $[\mathrm{OH^-}] = 2s = 2 \times 1.65 \times 10^{-4} = \mathbf{3.3 \times 10^{-4}}$ mol/L
> (c) $\mathrm{pOH} = -\log(3.30 \times 10^{-4}) = 3.48$; $\mathrm{pH} = 14.00 - 3.48 = \mathbf{10.52}$

4. $\mathrm{Ag_2CrO_4}$.

> $[\mathrm{Ag^+}] = 2s$, $[\mathrm{CrO_4^{2-}}] = s$: $K_{sp} = (2s)^2 s = 4s^3 = 4 \times (6.5 \times 10^{-5})^3 = \mathbf{1.1 \times 10^{-12}}$

5. Comparing.

> AgCl: $s = \sqrt{1.8 \times 10^{-10}} = 1.3 \times 10^{-5}$ mol/L. $\mathrm{Ag_2CrO_4}$: $s = 6.5 \times 10^{-5}$ mol/L, **five times larger**. Their Ksp formulas differ (s^2 versus $4s^3$), so Ksp values can only be compared directly for salts with the same ion ratio.

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C. Common ion and precipitation

6. CaF_2 in NaF.

> $[F^-] \approx 0.010$ mol/L; $K_{sp} = s(0.010)^2$, so $s = \dfrac{3.9 \times 10^{-11}}{1.0 \times 10^{-4}} = \mathbf{3.9 \times 10^{-7}}$ mol/L, about **550 times less** than in pure water.

7. Mixing.

> Total volume 50.0 mL.
> $[Ba^{2+}] = 0.010\ \text{mol/L} \times \dfrac{10.0\ \text{mL}}{50.0\ \text{mL}} = 2.0 \times 10^{-3}\ \text{mol/L}$
> $[SO_4{}^{2-}] = 0.0050\ \text{mol/L} \times \dfrac{40.0\ \text{mL}}{50.0\ \text{mL}} = 4.0 \times 10^{-3}\ \text{mol/L}$
> $Q = (2.0 \times 10^{-3})(4.0 \times 10^{-3}) = 8.0 \times 10^{-6}$, far greater than $K_{sp} = 1.1 \times 10^{-10}$: **$BaSO_4$ precipitates**.

D. Challenge

8. Milk of magnesia.

> $Mg(OH)_2\,(s) \rightleftharpoons Mg^{2+} + 2\,OH^-$. Stomach acid supplies H^+, which removes OH^- by forming water ($H^+ + OH^- \longrightarrow H_2O$). By Le Chatelier's principle, the equilibrium shifts to the **right** to replace the OH^-, so more $Mg(OH)_2$ dissolves, neutralizing the acid.

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